Performance Analysis of Football Players Via a Combination of Principal Component Analysis and K-means Clustering

To no one's surprise, football is one of the most popular sports in the world, bringing people from different ends of the earth together. This sport is one that is even able to catch the attention of many off the field as people analyse players, bet on teams, play football video games and even establish fan clubs to support their favourite teams. As technology advances, the data collection and analysis process of football player statistics improve over time, enabling us to evaluate football performance in a much more efficient and accurate manner. In the competitive world of football that is continuously evolving, it is of utmost importance for players, coaches, managers and other relevant parties to understand player performance as it helps in strategic planning, talent scouting and so forth for the betterment of a team.

Following the availability of comprehensive player statistics, we can uncover valuable player performance patterns in any manner that we desire. As such, this short study focuses on the performance of football players, specifically forward players, via the K-means clustering method with the incorporation of Principal Component Analysis (PCA). This study aims to provide an overview of how forward football players from different squads and leagues can be clustered into different groups based on their respective performance metrics, besides discovering the performance metric that is the biggest driver of the cluster separation. This study emphasises the usage of clustering, which looks to find homogeneous subgroups among the football players, particularly through K-means clustering, one of the most commonly used and simple clustering techniques for unsupervised learning tasks. Here, K-means is combined with PCA, an unsupervised dimension reduction technique, which aims to discover a low-dimensional representation of the data that can adequately capture the variance. Worth noting, it is typical to project data using PCA onto a lower-dimensional subspace, where K-means is subsequently employed (Zha et al. 2002). These techniques are explained further in the upcoming sections. By conducting this performance analysis, we can more fully comprehend the underlying patterns and traits that make up the population of football players.

Data Set

In generating clusters to group forward football players, the “2022-2023 Football Player Stats” data set is utilised. It contains 2022-2023 football player statistics per 90 minutes, where players from the Premier League, Ligue 1, Bundesliga, Serie A and La Liga leagues are included. This data set consists of 2689 observations and 124 variables, namely that of player descriptions or personal details and various types of performance metrics. The data set was collected by Vivo Vinco, who obtained it from Football Reference (n.d.). The full description of the data set’s variables can be obtained from Kaggle (Vinco 2023). The diverse array of performance metrics in this data set can provide us with a comprehensive understanding of player achievements and contributions on the field.

Techniques and General Procedure

This section explains the techniques used to cluster the forward football players, besides explaining the process step by step over various subsections. The steps taken in this study are similar to that of Elitsa Kaloyanova’s write up in the 365 Data Science website (Kaloyanova 2021).

Data Pre-processing

As with any analysis, the data set first needs to be read in, particularly into Jupyter in this study. The data cleaning process is carried out by removing any observations with missing values. Apparently, only one observation has a missing value for the *Nation* column, leaving us with 2688 observations and 124 columns. In the performance analysis carried out, we focus only on players with the “FW” or forward position and thus filter out all the other positions. In football, there are a few types of forwards, namely centre forward, striker, second striker, winger and False 9. Forward players are responsible for scoring goals for their team most of the time as they are attacking players that play closest to the goal of the opposition. They are required to have scoring, shooting, header and dribbling skills. Forwards players are chosen in this analysis since it is crucial to assess their performances, as their strengths and weaknesses can be leveraged by coaches to plan their match strategy. They are important players as a team’s style of playing can revolve around their attacking prowess and capabilities, besides evaluating their efficiency in front of the goal post. The identification of high performing forwards is also essential during the club transfer season as teams compete for the best forwards across clubs.

Following the filtering for only forward players, we are left with 408 observations altogether. The next step is to remove redundant and unnecessary variables that are not needed for our performance analysis that solely focuses on performance metrics. It is worth noting that in this performance analysis, personal details of players are deemed unnecessary as they are irrelevant and do not contribute to a player's performance. Besides that, the removal of variables helps in reducing the data set's dimensionality, apart from improving the analysis's effectiveness and manageability. The reasons for the removal of 26 variables are listed in Table 1, leaving us with 98 variables and 408 observations for analysis.

Table 1 Removal of redundant or unnecessary variables from the data set

Variable	**Reason for Removal**
Rk Player Nation Squad Comp Age Born	Irrelevant to the performance analysis carried out
Pos	Contains only the "FW" position as the others were filtered out
Min	Provides the same information as *90s* but in different units. *90s* is retained as it is a standardised measure that is not influenced by the number of matches a player has played in, thus enabling comparisons between players with different number of matches played.
SoT%	Can be derived from *SoT* and *Shots*.
G/Sh	Can be derived from *Goals* and *Shots*.
G/SoT	Can be derived from *Goals* and *SoT*.
PasTotCmp%	Can be derived from *PasTotCmp* and *PasTotAtt*.
PasShoCmp%	Can be derived from *PasShoCmp* and *PasShoAtt*.
PasMedCmp%	Can be derived from *PasMedCmp* and *PasMedAtt*.
PasLonCmp%	Can be derived from *PasLonCmp* and *PasLonAtt*.
PasAtt	Identical to *PasTotAtt*
PasCmp	Identical to *PasTotCmp*
Crs	Identical to *PasCrs*
TklW	Identical to *TklWon*
Tkl+Int	Can be obtained from *Tkl* and *Int*

TouLive	Identical to *Touches*
AerWon%	Can be derived from *AerWon* and *AerLost*
TklDri%	Can be derived from *TklDri* and *TklDriAtt*
ToSuc%	Can be derived from *ToSuc* and *ToAtt*
ToTkl%	Can be derived from *ToTkl* and *ToAtt*

Principal Component Analysis

PCA is a dimensionality reduction technique that can be employed prior to clustering to reduce the number of variables, especially for large data sets. PCA looks for the best subspace that can account for the majority of the data's variance and turns the original features into what are referred to as components. These components are essentially new variables derived from the original variables in the data set. At the end of the PCA analysis, we strive to select as few components as we can while retaining as much of the original data as possible. The first principal component represents the direction of the most variation, and each following component explains less variation than the one before it. Thus, instead of utilising the original variables, they are transformed and then fit to the model, where the transformed variables are usually smaller in number than they originally were. The principal components, $z_1, z_2, \ldots, z_M$, represent $M < p$ linear combinations of the original p variables. As such, the m^{th} principal component is represented by:

$$z_m = \sum_{j=1}^{p} \phi_{jm} X_j$$

where ϕ_{jm} is the loading of the j^{th} original variable, X_j. Loadings represent the contribution of each variable to the principal component. The PCA algorithm ensures that all of the principal components are orthogonal or uncorrelated to one another.

Before performing dimensionality reduction using PCA, the data set needs to be standardised using `scale()`. This is because the variables all have different units or scales, where much bigger weights will be put on variables with larger values if not normalised beforehand. This standardisation step will ensure equal importance of each variable during clustering. Next, PCA is employed using the `PCA()` and `fit_transform()` functions on the scaled data set. PCA will create as many principal components as there are variables in the data set, 98 in our case, if not set beforehand. PCA has managed to transform the data linearly, producing 98 new components, which explain 100% of variability in the data. Some of the components include a substantial amount of variation, while others contain essentially none. From Figure 1 below, we can see that the first component explains about 20.66% of the data variability, the second component about 6.96% and so on.

```
np.round(pca.explained_variance_ratio_, decimals=4)*100
```

```
array([2.066e+01, 6.960e+00, 4.550e+00, 4.200e+00, 4.060e+00, 3.630e+00,
       2.880e+00, 2.630e+00, 2.510e+00, 2.400e+00, 2.200e+00, 1.960e+00,
       1.920e+00, 1.850e+00, 1.760e+00, 1.690e+00, 1.530e+00, 1.500e+00,
       1.430e+00, 1.290e+00, 1.250e+00, 1.220e+00, 1.200e+00, 1.130e+00,
       1.060e+00, 1.060e+00, 1.030e+00, 1.010e+00, 9.700e-01, 9.200e-01,
       8.900e-01, 8.800e-01, 8.500e-01, 8.100e-01, 7.800e-01, 7.500e-01,
       7.300e-01, 7.200e-01, 6.700e-01, 6.400e-01, 6.200e-01, 6.000e-01,
       5.600e-01, 5.200e-01, 5.000e-01, 4.800e-01, 4.400e-01, 4.100e-01,
       3.800e-01, 3.700e-01, 3.500e-01, 3.500e-01, 3.000e-01, 2.900e-01,
       2.700e-01, 2.600e-01, 2.500e-01, 2.400e-01, 2.200e-01, 2.000e-01,
       1.900e-01, 1.800e-01, 1.600e-01, 1.600e-01, 1.500e-01, 1.400e-01,
       1.300e-01, 1.300e-01, 1.100e-01, 1.100e-01, 9.000e-02, 9.000e-02,
       8.000e-02, 7.000e-02, 6.000e-02, 5.000e-02, 5.000e-02, 5.000e-02,
       3.000e-02, 3.000e-02, 2.000e-02, 2.000e-02, 1.000e-02, 1.000e-02,
       1.000e-02, 1.000e-02, 1.000e-02, 1.000e-02, 1.000e-02, 0.000e+00,
       0.000e+00, 0.000e+00, 0.000e+00, 0.000e+00, 0.000e+00, 0.000e+00,
       0.000e+00, 0.000e+00])
```

Figure 1 Explained variance values of each principal component

Since the components are ordered in decreasing order of variance explained, we need to select the first few components that preserve a substantial amount of information. The cumulative explained variance ratio plot for PCA components in Figure 2 can help us determine how many components to retain in the dimensionality reduction. A general rule of thumb would be to keep about 80% of the variance. From the plot, we can see that the first 28 components explain at least 80% of the variance, which could significantly reduce the variables. The next step is to refit the PCA with 28 components by setting `PCA(n_components = 28)` and using `fit_transform()` on the scaled data once again. These steps will enable us to perform K-means clustering on the principal component scores instead of on the original variables.

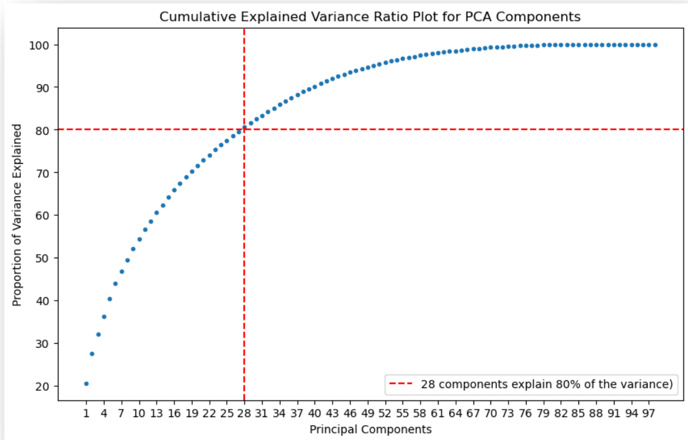

Figure 2 Cumulative explained variance ratio plot for PCA components

K-means Clustering

Clustering is a method that aims to group observations into clusters that are homogenous within themselves. K-means clustering is a clustering technique that requires the number of clusters, usually denoted as K, to be determined before the analysis is carried out. The union of every cluster contains each and every one of the observations (all 408 observations in our case), where each observation belongs to one and only one cluster. K-means clustering is a hard clustering method such that an observation belongs entirely to a cluster or does not at all. The within-cluster variation, $W(C_k)$, should be as minimal as possible for a clustering to be deemed good. This technique aims to obtain partitions such that the sum of the within-cluster variation over each of the K clusters is as minimal as possible, portrayed mathematically as follows:

$$\underset{C_1, \dots, C_k}{min} \left\{ \sum_{k=1}^{K} W(C_k) \right\}$$

Most commonly, the within-cluster variation concept is defined by the Euclidean distance as follows:

$$W(C_k) = \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2$$

such that $|C_k|$ represents the number of observations in the k^{th} cluster.

In conducting K-means clustering, the first step is to determine the number of clusters. Each observation is then randomly assigned to one of the clusters, before computing the centroid of each cluster, which is essentially the mean of the observation values in that cluster. Using Euclidean distance as mentioned earlier and based on whichever centroid is the nearest, all the observations are redistributed. The new centroid for each cluster is computed once again and the process is repeated over and over again until the determined number of maximum iterations is achieved or until the centroids stay constant.

For this study, the `KMeans()` function from the sklearn.cluster library is used to conduct clustering. It is essential to note that the `n_init` argument is to set the number of times initial random configurations will be used to perform K-means clustering, with only the best results (the assignments producing the lowest total variance) being reported in the end result. For this study, the number of random initial configurations is set to 50. The `random_state` on the other hand is set to 1 to ensure the reproducibility of results. The total within-cluster sum of squares that K-means aims to minimise can be determined through the `.intertia_` function. There are various approaches to determine the optimal number of clusters to be implemented in K-means clustering. Two of the approaches utilised in this study are the Elbow method and Silhouette analysis. The Elbow method involves a plot of total within-cluster sum of squares against the number of clusters. Here, the optimal number of clusters is chosen to be the point at which an "elbow" is spotted, that is the area before the elbow would be sharply decreasing whereas after it, it would be considerably smoother (Kaloyanova 2023). The Silhouette analysis on the other hand, is a similarity index of data points within a cluster and between clusters (Banerji 2021). Silhouette scores can range from -1 to +1, with higher scores indicating better clustering performance. The Silhouette coefficient for a particular observation, i, is defined mathematically as follows:

$$S(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where $a(i)$ is the average distance between i and every other data point in the cluster to which i belongs, while $b(i)$ is the average distance between i and every cluster to which it does not belong. The Silhouette score is then calculated by averaging $S(i)$ for every cluster. The results of the Elbow method and Silhouette analysis for up to 15 clusters are shown in Figure 3, followed by their discussions below the figure.

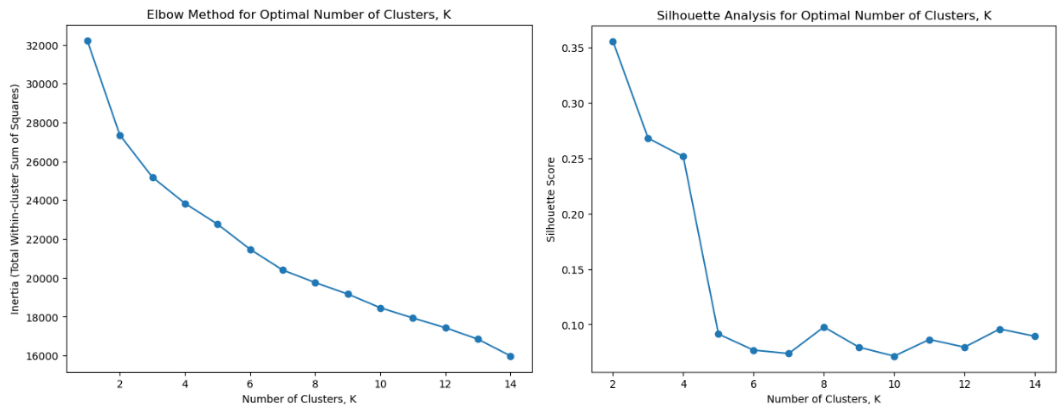

Figure 3 Elbow method and Silhouette analysis plots to determine the optimal number of clusters for K-means clustering

From Figure 3 above, the Silhouette analysis clearly indicates 2 as the optimal number of clusters, seeing as it has the highest silhouette score. However, based on our specific objective of clustering players based on their performances, two clusters might not be such a good choice. Increasing the number of clusters to the next highest silhouette score, three clusters, would be a much better option. Two clusters would be too simplistic because for some variables it would have one cluster with high scores and the other cluster with much lower scores. With more than two clusters, the clusters would be able to capture player performances more appropriately, leading to more interpretable insights. Furthermore, the Elbow method plot does not show a clear indication of where the "elbow" is. Nevertheless, if observed closely, we can see that until three clusters there is a steep decline in inertia values, followed by a more constant decline after that. Following this, we use the `KMeans()` function with an initialiser of 50 and random state of 1 on three clusters. The model is then fitted on the principal component scores from earlier.

<u>Cluster Visualisation and Bar Plots of Offensive, Defensive and Disciplinary Metrics</u>

Once the clustering algorithm has been implemented, the three clusters are to be visualised on a 2D plane. Since PCA determined the most important components in order, the first two components are utilised as the axes of the plot. To observer the clusters, a scatter plot of the second principal component against the first principal component is plotted, with the observations coloured according to their respective clusters. One of the good things about conducting PCA before clustering is that nearly all of the data can be visibly segregated as will be seen in the following subsections.

Moreover, 12 of the variables are selected and grouped into offensive, defensive and disciplinary metrics as follows:

Offensive metrics: *Goals, SoT, ToSuc, Assists*

Defensive metrics: *TklWon, Int, AerWon, Clr*

Disciplinary metrics: *Fls, CrdY, 2CrdY, CrdR*

Only four variables that are deemed important based on judgement are chosen for each type of metric to provide simplicity in this case. These performance metrics are averaged across their clusters and plotted as a bar plot to obtain a general overview of the skill set players in each cluster possesses. The comparisons made between clusters can highlight the unique patterns and traits of each cluster, allowing us to determine which clusters excel in particular domains and which clusters may need improvements. The results are plotted in Figure 5 to aid in the interpretation of Figure 4.

Most Distinctive Variable Between Clusters

In this subsection, the variable that contributes most to the separation of the clusters is determined using two different approaches, namely through PCA loadings and the comparison of means across clusters, leading to two different results. By visualising the most distinctive variable between clusters, it is easier to comprehend the variables that set the clusters apart from one another more thoroughly. It aids mainly to explain the significance of each cluster and what distinguishes them in terms of player performance. Besides that, it helps us to pinpoint particular performance metrics that significantly influence the clustering patterns, which might have not been apparent through the cluster plot alone.

First off, we could compute PCA loadings to identify the most distinctive variable between clusters, as they detail the contributions made by each original variable to the principal components. It can be said that a variable's impact on a principal component is greater when it has a higher loading. As such, the `.components_.T` function is used to obtain the PCA loadings for all 28 principal components across each of the 98 original variables. The mean absolute loading for each variable across all principal components are then calculated, where the variable with the highest mean absolute loading is determined as the most distinctive variable according to this method.

The contribution of a certain variable to cluster separation can also be determined via the computation of centroids or mean of variables according to their respective clusters. This concept is heavily connected to K-Means clustering itself. Since the mean is the average of the observation values, it is useful in summarising and providing an overview of a cluster's characteristics, helping us to understand how the clusters differ according to various variables. The variable that significantly differs between clusters can be found by comparing the means, with the most distinctive variable in separating clusters being the one that has the biggest variance of mean amongst clusters. This method could do a good job if the clusters are distinctly separated from one another. To carry out the method, the process starts with grouping the data set by clusters and finding the mean of each variable by their respective clusters. Then, compare the means for each variable across clusters, where the variable displaying the biggest variance is deemed as the most distinctive variable according to this method. Great differences in means of a variable and a statistically significant difference between clusters, indicates that the variable contributes a big deal to the segmentation of observations into clusters.

Once the most distinctive variables using both methods have been identified, Analysis of Variance (ANOVA) is performed to evaluate the significance of that particular variable. It ascertains the statistical significance of the mean differences between the clusters to confirm the validity of the variable for cluster separation. In our case, ANOVA compares the means of a variable across three clusters, with null and alternative hypotheses as follows:

$$H_0: \mu_1 = \mu_2 = \mu_3$$

$$H_{1:}\text{At least one cluster's mean is different from the rest}$$

Here, we set out significance level to 0.05. If the p-value obtained is less than 0.05, the null hypothesis is rejected and it can be said that the most distinctive variable is significant. Conversely, a p-value that is greater than 0.05 indicates that the most distinctive variable is not significant since its mean is the same across all clusters. This could portray that the values of that particular variable across all three clusters are nearly the same and does not drive a distinct clustering. Once ANOVA has been completed, a boxplot of the most distinctive variable is plotted across clusters to visualise its distribution in the three distinct clusters and aid in our comprehension of each cluster's characteristics. Box plots without any overlapping could portray the importance of the variable in influencing cluster separation. This way, we can observe how the forward players are grouped based on one particular performance metric and uncover underlying patterns.

Results and Discussion

Cluster Visualisation and Bar Plots of Offensive, Defensive and Disciplinary Metrics

In this study, the objective is to conduct a performance analysis of forward players by carrying out K-means clustering on principal component scores. The scatter plot in Figure 4 displays the PCA's two most important principal components, namely the first and second principal component, which together account for about 80% of the data set's total variance. Even though there is a slight overlap between Cluster 1 and Cluster 2, the three clusters are seen to be successfully segregated into separate groups as a whole, based on player performance metrics. This could indicate that the players in Cluster 1 and Cluster 2 respectively, have the same performance characteristics within themselves, while still retaining some clear distinctions. The "X" markers on the other hand, represent the centroids for each cluster, which is the average values of the principal components 1 and 2 according to cluster. The centroids are located at nearly the same level for the principal component 2 axis, but vary largely for the principal component 1 axis, with Cluster 3 taking the lead, followed far behind by Cluster 1 and then Cluster 2. Loadings that are positive in value portray a positive relationship with a particular principal component and vice versa. A higher magnitude of loadings value indicates a stronger association. Based on the eight offensive and defensive metrics from earlier, the analysis shows that for the offensive metrics, principal component 1 is the more dominant component, whereas principal component 2 is more dominant for the defensive metrics. The results of Cluster 3 having the highest centroid value for principal component 1, followed by Cluster 1 and Cluster 2 is in tandem with what is discussed in the next figure.

By looking at all the points, we might notice some of them straying far away from their clusters, especially for some players in Cluster 1. These players are considered as outliers that may be reflective of their impressive performance metrics, causing them to standout among the rest. The area in which they exhibit exceptional performances could be in the four defensive aspects listed before (among many other aspects not analysed), as they have high values in the principal component 2 axis.

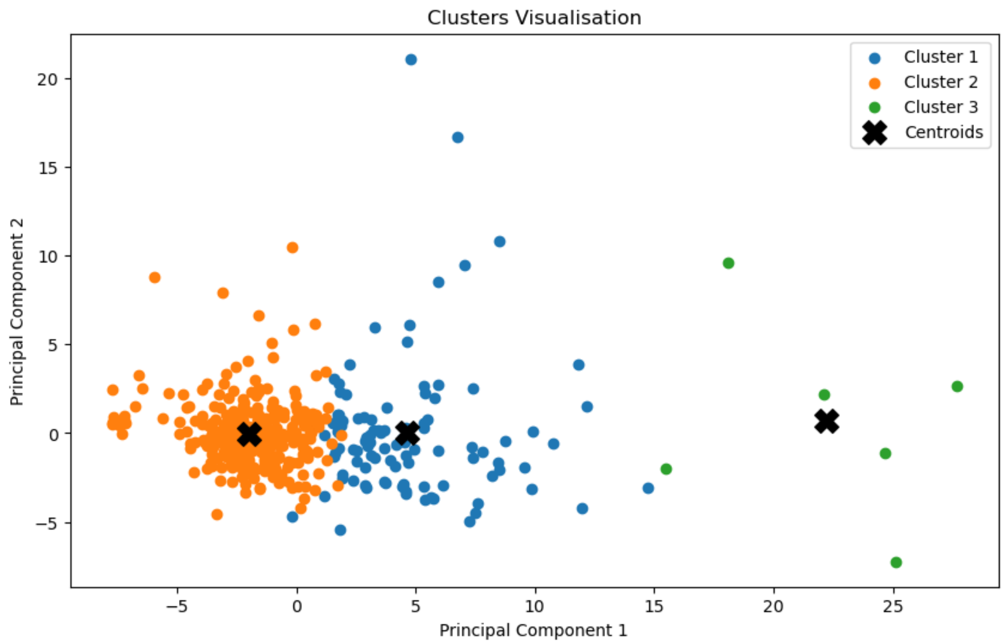

Figure 4 Clustering visualisation by conducting K-means clustering on principal component scores

Notably from Figure 4, Cluster 2 has the greatest number of forward players, followed by Cluster 1. There are only a few players being placed in Cluster 3, indicating quite a unique bunch of forward players with more distinct traits. From the results obtained in Figure 5, which are based on four variables each, we note that forward players from Cluster 3 have the best offensive skill set. As mentioned earlier, these few unique players are able to display excellent skills in terms of scoring goals, taking shots on target, taking on their opposition team successfully and providing assists to their teammates. This is exactly what a forward player is most responsible for, indicating that the players from Cluster 3 are essentially the cream of the crop of forward players in this football season. The six players with the best offensive skill set according to our analysis are Josip Brekalo (Fiorentina, Serie A), Arnaut Danjuma (Tottenham, Premier League), Siriki Dembélé (Auxerre, Ligue 1), Malcom Abdulai Ares Djalo (Athletic Club, La Liga), Memphis (Atlético Madrid, La Liga) and Adam Ounas (Napoli, Serie A). However, the players from Cluster 3 have the lowest average defensive metrics, which could just mean that these players truly focus on their offensive skill set, playing mostly in the area of their opponent's pitch and goal post instead of their own.

Even though this analysis is concerned about forward players, it seems that players from Cluster 2 have the best defensive skill set based on their abilities to succeed during tackles, intercept passes, win aerial duels and make clearances during a match. Cluster 1 comes in second, not far behind for both the offensive and defensive metrics, indicating that this cluster consists of forward players who have an average performance in terms of offense and defence. It is also worth noting that players from Cluster 1 have the lowest disciplinary metrics average, indicating that they are the most disciplined forward players among the three groups as they receive less fouls, yellow cards and red cards. Players from Cluster 2 on the other hand, have the highest average disciplinary metrics, which could be attributed to their more aggressive playing style during defence, in which physicality often takes place. It is logical for them to have more fouls and yellow/red cards than those who are more focused on offense. It can be said that the K-means clustering algorithm did a good job in capturing the traits of player performances, more so since each cluster's performances can be interpreted with ease.

Figure 5 Bar plots of average offensive, defensive and disciplinary metrics across clusters

Most Distinctive Variable Based on PCA Loadings

Based on the analysis carried out, the variable with the highest mean absolute loading is *CarDis*, and thus identified as the most distinctive variable among clusters based on the PCA loadings method explained earlier. *CarDis* is the number of times a player loses control of the ball after being tackled by an opposing player, where higher values could indicate a worse performance. However, when carrying out ANOVA to test the statistical significance of the *CarDis* mean differences between the clusters, it was found that the $p - value = 0.5095 > 0.05$. Therefore, the null hypothesis has been failed to be rejected and there is insufficient evidence to support that the *CarDis* means significantly differ among the clusters. This could indicate that *CarDis* is not the most suitable variable to use in explaining the differences between the three clusters obtained. Even by looking at the distribution of *CarDis* values across clusters from the box plot in Figure 6, it is noticeable that the medians of all three clusters are nearly the same, especially that of the first and second cluster. As such, determining the most distinctive variable to explain the differences between clusters using PCA loadings is not the most appropriate method to use in our case. It is important to note that while *CarDis* may contribute the most to the formation of the principal components, it is not necessarily driving the difference between clusters.

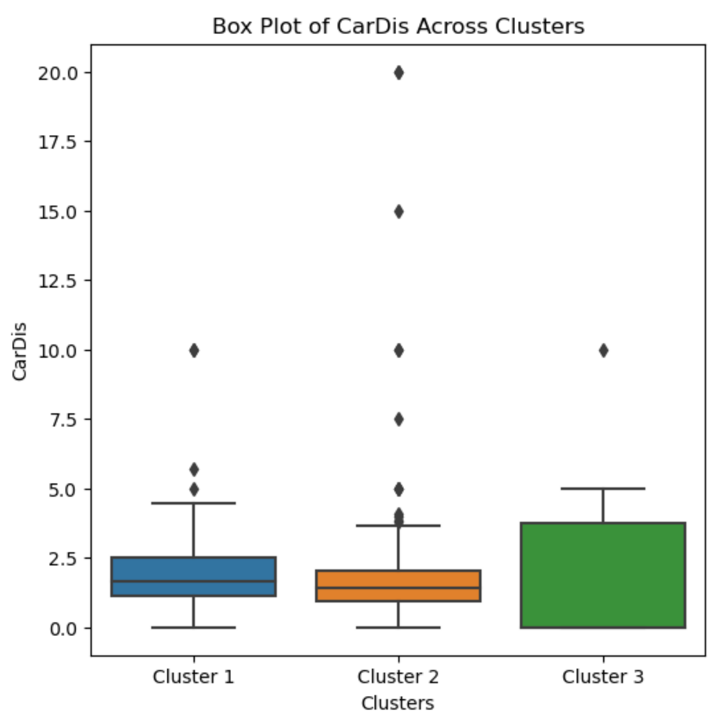

Figure 6 Box plot displaying the distribution of *CarDis* across all three clusters

Most Distinctive Variable Based on Mean of Variables Across Clusters

Following the fact that the PCA loadings method is not suitable to determine the most distinctive variable, the variable with the highest variance of its mean across clusters is observed instead. This is sure to bring about a significant difference across clusters as it has the biggest variance. Based on this method, the most distinctive variable is identified to be *PasTotDist*, which is the total distance that completed passes have travelled in any direction, in yards. To validate that the mean of *PasTotDist* across all three clusters is statistically significant, the ANOVA outputs $p - value = 4.761 \times 10^{-98} < 0.05$. Thus, the null hypothesis is rejected and the mean of *PasTotDis* is statistically different across clusters, further indicating that this variable is essential in differentiating and segmenting the clusters. As such, it is important to understand what this variable is all about. It offers important details regarding a player's potential for moving the ball around the field efficiently and contribution to playmaking that benefits the team as a whole. Players whose completed passes can cover further distances, possess great passing and playmaking skills as they continuously strategise and orchestrate their movements throughout the match. This way, their team can maintain possession of the ball and control the match, besides opening up chances to score goals through their passes. Smaller covered distances on the other hand, could be an indication of a more unique ball distribution strategy, or in a worse case scenario, lesser completed passes.

The box plot in Figure 7 portrays the distribution of *PasTotDist* values across the three clusters to gain further insights. Notably, the text in green are the players' names and *PasTotDist* value for the greatest value of that cluster, whereas the red is for the minimum value. At a glance, it is very obvious that players from Cluster 3 have much higher values of *PasTotDist*, with its median at around 1000 yards. The smallest value within the lower inner fence of Cluster 3 is higher than the largest value within the upper inner fence of the other two clusters. This shows that players from this cluster are just absolutely extraordinary in their complete passes distance with much higher ground covered. This is especially true for Adam Ounas, a French-Algerian player, with a total distance covered of 1485 yards for his completed passes, which is well above the median. This showcases his exceptional accuracy in long passing. The range of values is not that large as well, indicating that the performance of players in Cluster 3 based on *PasTotDis* does not vary that much, showing consistency in their great performance.

Here, players from Cluster 2 have the lowest values of *PasTotDist* among all three clusters, with Carlos Álvarez having a *PasTotDist* of 0. This could mean that he has not had any completed passes during the matches he played. The smaller values of *PasTotDist* for Cluster 2 does not simply mean that the players have bad performances, as it could also indicate a different playing style. For example, these forward players might be more focused on short passes that cover less ground or specifically focus more on a certain area of the field. More in depth analysis would be required to understand why their values are as such and how it relates to their strategies. The outlier in Cluster 2 could indicate players with more precise passes as they complete their passes but at shorter distances. This could be their strategy in controlling the ball possession, besides executing quick passes that could disrupt their opponents' game. The median of Cluster 1 is closer to that of Cluster 2 than it is to Cluster 3, although its outliers especially Diego Collado, could belong to Cluster 3 due to his high *PasTotDist* value of 1110. He could however be in Cluster 1 due to the other various performance metrics utilised in our analysis. Cluster 1 also has the biggest range, indicating that players from this cluster have more varying performances in terms of *PasTotDist*.

Also noticeable, all the players with the greatest *PasTotDist* values within their respective clusters are classified as outliers. Diego Collado, Pierre-Emerick Aubameyang and Adam Ounas all surpass the typical range of total distance for completed passes for players within their clusters. They showcase performances that are either extraordinary or just have different playing styles than that of the other players.

The results can be summarised by concluding that players from Cluster 3 play an essential role in creating chances for goal scoring from afar, besides enabling a switch of play as they make long passes to their teammates. Conversely, players from Cluster 2 are more rapid in their executions as they make short and accurate passes, which could indirectly control the tempo of the game. Additionally, the performances of players from Cluster 1 lie somewhere in the middle of these two clusters. As the differences between clusters are clearly distinguishable, *PasTotDist* is an essential metric in exploring the differences in player performance. It is important to keep in mind that there are many other factors influencing the clustering and this variable is only one of the most distinctive drivers of the separation. The results can help coaches in strategising their team's gameplay based on this important variable that separates players from one another, as it helps them to understand their players' capabilities.

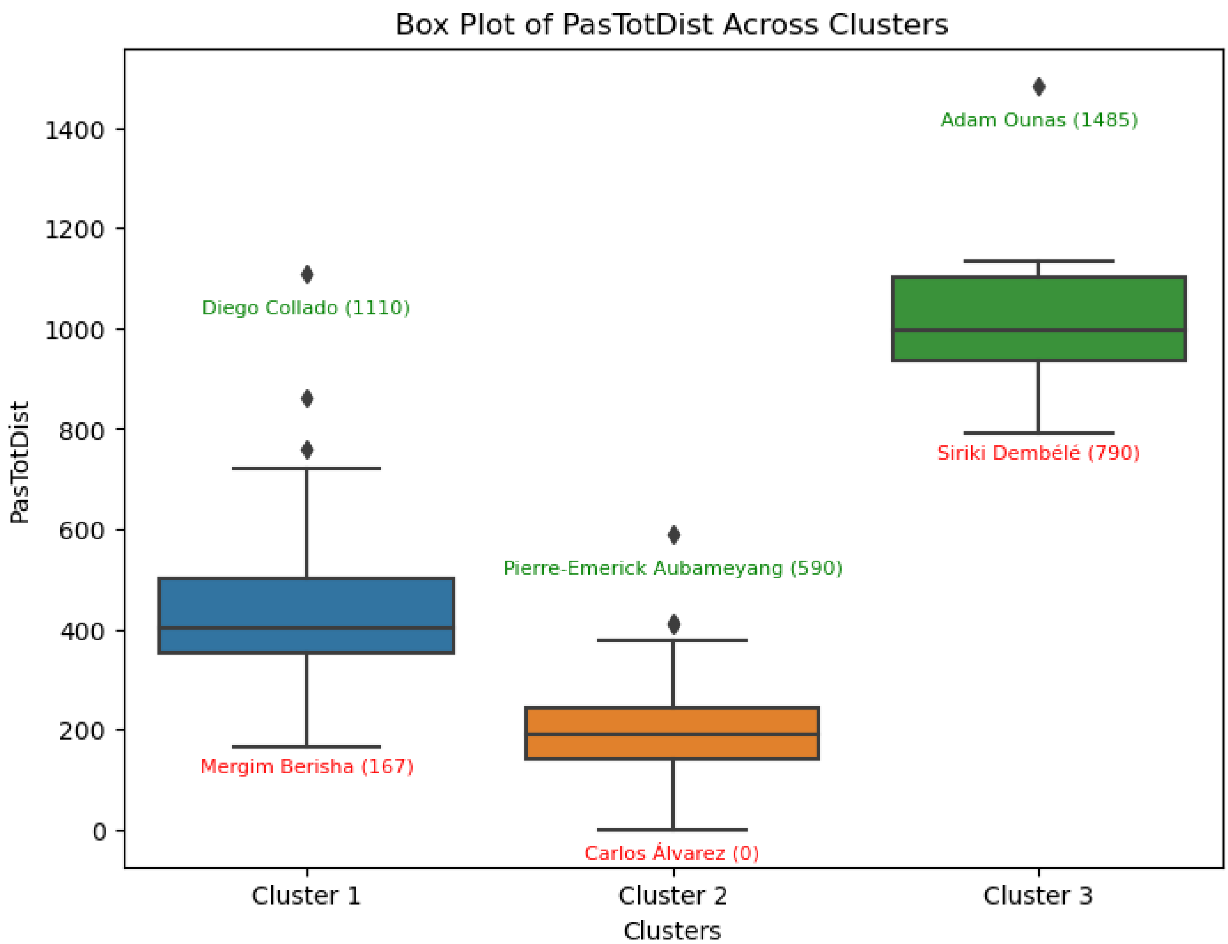

Figure 7 Box plot displaying the distribution of *PasTotDis* across all three clusters

Conclusion

The combination of PCA and K-means clustering has deepened our understanding of how the performance metrics of forward football players are distinct from one another. This can further aid in the comprehension of a player's roles, specific skill set, playing style and contributions to their teams based on their respective clusters. We have managed to obtain many valuable insights from the analyses carried out on the three clusters obtained. Firstly, it was found that the six players from Cluster 3 are the best forward players as they depict the highest values for the offensive metrics. Players from Cluster 2 on the other hand, have better defence skills even though they are forward players, besides having the most disciplinary issues. Players from Cluster 1 have a more balanced offensive and defensive skill set. The centroids of the clusters also help in understanding the skill set of the players based on the dominant principal components.

Besides that, *PasTotDist* was determined as the most distinctive variable that drives the separation of the three clusters as it has the biggest variance in cluster means between all the variables. It was found that players in Cluster 3 have the longest passing capabilities, followed by players from Cluster 1 and Cluster 2. The players in Cluster 2 specifically, have a unique style of playing in that their passes are quicker and shorter with a fast-paced game. This variable is definitely an important aspect in assessing a player's performance as they vary widely between players. Coaches and managers can emphasise on this specific aspect based on which cluster their player belongs to, in order to improve the overall performance of the team. All the results obtained can assist the management of a team's players based on their respective characteristics.

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